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Evaluation of the Particle Growth of Amorphous Calcium Carbonate in Water by Means of the Porod Invariant from SAXS

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A time-resolved SAXS study has been carried out on the formation of amorphous calcium carbonate from supersaturated aqueous solutions at an initial concentration of 5 mmol/L $CaCO_3$. Particle formation was induced by mixing equal volumes of equinormal $CaCl_2$ and Na_2CO_3 solutions with a stopped-flow device installed at the SAXS beamline. The resulting scattering curves were analyzed without any model assumption with respect to the particle shape. The analysis is based on the intercept of the scattering curve, its initial slope, and the Porod invariant. These parameters give access to the average particle mass, the average particle size, and the mass concentration of the particles, respectively. The evolution of particle mass and concentration with time gives access to the trend in the particle number density. The size and mass values were found to be correlated by characteristic exponents. Two different mass values can be used for this correlation: direct use of the intercept of the scattering curve or alternatively a ratio of this intercept with the corresponding Porod invariant. The resulting exponents depend on the particle growth mechanism. These exponents, together with the evolution of the number density, are capable of discriminating between a monomer-addition mechanism and a particle−particle coagulation mechanism as two alternative building mechanisms for the resulting amorphous $CaCO_3$ nanoparticles. A detailed description of the data analysis and its merit in establishing a particle growth mechanism is presented.

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Introduction

The formation of calcium carbonate from supersaturated solution is a ubiquitous chemical process both in nature and in technical applications. In nature, this process is an important aspect of biomineralization and leads to high-performance materials such as those in the exoskeleton of molluscs or the spines of sea urchins.[1] Unlike with biomineralization, calcium carbonate formation frequently turns into a highly undesirable side reaction once hard water is involved in technical processes such as household laundry or the circulation of hot water in pipe systems.[2,3] Therefore, a detailed understanding of calcium carbonate formation is highly desirable because it may serve as a source of inspiration for the development of new high-performance materials or it may help to better control or even prohibit it in technical applications.

At least four different polymorphs exist for solid $CaCO_3$, including amorphous calcium carbonate (ACC), vaterite, aragonite, and calcite. Under ambient conditions, the stability of the crystalline polymorphs increases along the sequence vaterite, aragonite, and calcite. Thermodynamically, the least stable modification is ACC.[4,5] The trend in solubility of the modifications varies inversely with the respective thermodynamic stability of the modification. This increasing order of metastability observed for a series of polymorphs with comparable stability within a given material follows the order of decreasing surface energy, which usually decreases with decreasing density. The trend in metastability is caused by an interplay of the surface energy and bulk energy and appears below a certain particle size. As a consequence, the stable modification is approached via the Ostwald−Volmer rule by passing intermediate phases with decreasing metastability.[6]

Rieger et al. could clearly demonstrate that the initially formed ACC gradually transforms into stable calcite crystals via the state of dissolved ions in equilibrium.[7] The disappearing ACC exhibited a granular structure with constituent ACC particles smaller than 100 nm. The addition of polyacrylate chains led to the stabilization of ACC nanoparticles, supporting the assumption of small precursor particles.

A considerable step forward in the understanding of the ACC formation was provided by Ballauff et al., who published a series of striking time-resolved small-angle X-ray scattering experiments.[8−10] Using a stopped-flow device, they succeeded in providing a series of captivating scattering curves with clear-cut evidence for spherically shaped particles with an extremely narrow size distribution. A straightforward application of model fits based on regular spheres with a finite size distribution gave evidence for a short period of nucleation that led to a constant number of growing particles. The $CaCO_3$ level covered a regime of 3.5 mM ≤ [$CaCO_3$] ≤ 4.5 mM. In all cases, an intermediate species with a constant particle size was achieved. The size of these metastable particles increased with the initial concentration of $CaCO_3$.[8] If this carbonate content was set to [$CaCO_3$] = 10 mM, then the particle distribution became broader. Results in the latter case indicated particles with a radius of 19 nm as basic constituents, which successively form larger aggregates.[10] The density of 1.62 g/cm^3, which is low compared to the densities of the three crystalline polymorphs, confirmed the

(1) Gower, L. B. *Chem. Rev.* **2008**, *108*, 4551–4627.
(2) Richter, F. H.; Winkler, E.; Baur, R. H. *J. Am. Oil Chem. Soc.* **1989**, *66*, 1666–1672.
(3) Rieger, J.; Hädicke, E.; Rau, I. U. *Tenside, Surfactants, Deterg.* **1997**, *34*, 430.
(4) Brecevic, L.; Nielsen, A. E. *J. Cryst. Growth* **1989**, *98*, 504–510.
(5) Clarkson, J. R.; Price, T. J.; Adams, C. J. *J. Chem. Soc., Faraday Trans.* **1992**, *88*, 243–249.
(6) Navrotsky, A. *Proc. Natl. Acad. Sci. U.S.A.* **2004**, *101*, 12096–12101.
(7) Rieger, J.; Thieme, J.; Schmidt, C. *Langmuir* **2000**, *16*, 8300–8305.
(8) Bolze, J.; Peng, B.; Dingenouts, N.; Panine, P.; Narayanan, T.; Ballauff, M. *Langmuir* **2002**, *18*, 8364–8369.
(9) Pontoni, D.; Bolze, J.; Dingenouts, N.; Narayanan, T.; Ballauff, M. *J. Phys. Chem. B* **2003**, *107*, 5123–5125.
(10) Bolze, J.; Pontoni, D.; Ballauff, M.; Narayanan, T.; Cölfen, H. *J. Colloid Interface Sci.* **2004**, *277*, 84–94.

amorphous nature of the particles.[8] If combined with wide-angle X-ray scattering, the same authors could unambiguously demonstrate that ACC is transformed into the thermodynamically stable calcite modification via dissolution and subsequent crystallization of the constituent ions,[9] in close agreement with the earlier findings by Rieger et al.[7]

Another interesting aspect of the mechanism of ACC formation was disclosed by Wegner et al.,[11] who provided evidence for the occurrence of a spinodal decomposition with a lower critical solution temperature of $T_c = 10$ °C. They applied an in situ generation of the CO_3^{2-} ions by means of the hydrolysis of dialkyl-carbonates. This ester hydrolysis requires a significant lag time to produce enough CO_3^{2-} ions and thus avoided inhomogeneities, which inevitably occur during the initial stages of any mixing of component solutions and may interfere with nucleation.

The latest piece in a puzzle was recently provided by Cölfen et al.[12] by means of a carefully performed analysis of the Ca^{2+} ion concentration while titrating a large excess of Na_2CO_3 solution with small increments of a $CaCl_2$ solution. The application of a Ca^{2+}-sensitive electrode detected only a constant fraction of the total number of Ca^{2+} ions added until nucleation sets in. This is compatible with the equilibrium between Ca^{2+} ions in the solution state and a cluster species of $CaCO_3$ during the whole prenucleation phase.

One issue that is still a matter of debate refers to the growth mechanism of ACC nanoparticles after nucleation and can be stated as follows: does the growth mechanism proceed with the coalescence of ACC particles with an ever increasing average size, or does growth follow a "monomer"-addition step? The present work will give a clear answer to this question without being able to specify the term monomer, which may correspond to constituent ions, molecular $CaCO_3$ units or clusters such as those identified by Cölfen et al.[12] The answer is provided by an analysis of the scattering curves performed without any model assumptions made about the shape of the particles. It simply takes advantage of the accessibility of the so-called Porod invariant.[13] The approach uses this invariant to establish particle mass values and number densities of the growing particles in solution and successively correlates the resulting mass and size parameters, thereby providing unambiguous indications for the particle-formation mechanism.

Experiments and Data Reduction

Materials. Calcium chloride hexahydrate $CaCl_2 \cdot 6H_2O$ (assay ≥99%) and sodium carbonate Na_2CO_3 (assay ≥99%) were purchased from Fluka (Buchs, CH). Sodium hydroxide (assay ≥99%) was supplied by Merck (Darmstadt, FRG). Doubly distilled water (conductivity < 0.1 μS/cm) served as the solvent.

Time-Resolved SAXS Experiments. Time-resolved SAXS experiments on the formation of ACC particles were performed at the ID02 beamline, ESRF, Grenoble, France. The SAXS instrument is equipped with a stopped-flow device that enables rapid mixing of the reactants.[14] The X-ray wavelength was 0.995 Å, and the sample-to-detector distance was 3 m. Two-dimensional SAXS patterns were recorded using a CCD-based detector (FReLoN) with single-photon sensitivity. The acquisition time for a single frame was either 20 or 50 ms. The formation of ACC particles in supersaturated $CaCO_3$ solutions was induced by mixing equal amounts of equimolar (10 mM) solutions of $CaCl_2$ and Na_2CO_3. The pH of the Na_2CO_3 solution was set to 10 with an uncertainty of ±0.1. The stopped-flow apparatus (SFM-3, by BioLogic, Pont de Claix, France) was equipped with three motorized syringes with controllable pumping speeds. The feeding tube from the mixer was connected to a capillary with a diameter of 1.5 mm. The capillary served as the scattering cell. The volume from the mixer to the capillary amounted to 20 μL, which created a deadtime of a few ms. Experiments were performed at flow rates of 1.6, 2, 2.5, and 3 mL/s. At least three runs have been carried out under each condition. The $CaCO_3$ content in all cases was 5 mM. Because this leads to a supersaturated solution of $CaCO_3$, we use the term "initial concentration" of $CaCO_3$ throughout the text whenever we indicate the amount of $CaCO_3$, which was mixed together at the onset of an ACC formation process. After each ACC formation run, the stopped-flow mixer and the capillary were cleaned with aqueous HCl and successively with distilled water. Experiments at all flow rates led to the same trends. However, it turned out to be difficult to always reproduce each kinetic detail of the particle formation, which may in part be due to slight pH variations. The pH has not only an inherent uncertainty but may be further affected by exposure of the sample to the atmosphere as we could not work under closed conditions. The paper presents a selection of four experiments denoted as WOP34, WOP35, WOP36 and WOP37, which were induced with a flow rate of 2 mL/s. The first two scattering curves of each kinetic experiment, recorded at 0.04 and 0.21 s had to be discarded as they did not display a clear Guinier region and therefore prohibited any significant data evaluation.

Processing of the Scattering Curves. Measured 2D time-resolved SAXS patterns were corrected for the dark level of the CCD and spatial inhomogeneities (flatfield) of the detector and normalized to an absolute scale using the recorded transmitted beam intensity and known pixel size of the detector and sample to detector distance.[14] The normalized 2D SAXS patterns were azimuthally averaged to obtain the 1D scattering profile. Appropriate background subtraction of the solvent results in a net differential scattering cross section of the solute denoted as $d\Sigma/d\Omega$. Hence, all original scattering curves are expressed as differential scattering cross sections ($d\Sigma/d\Omega$) in units of reciprocal millimeters versus momentum transfer q

$$q = \frac{4\pi}{\lambda}\sin(\theta/2) \tag{1}$$

with λ being the wavelength of the X-rays and θ being the scattering angle under which ($d\Sigma/d\Omega$) was observed. Figure 1A is an example of a time-resolved SAXS (TR-SAXS) experiment by means of a selection of scattering curves at variable times. The scattering curves can be subdivided into several factors according to

$$\left(\frac{d\Sigma}{d\Omega}\right) = I(q) = K_{SAXS} c M_w P(q)\, S(q) \tag{2}$$

Intercepts of $d\Sigma/d\Omega = I(q)$ are proportional to the solute concentration c in g/mL and to the apparent weight-average molar mass M_w of all solute species. The angular dependence of $I(q)$ is determined by a form factor $P(q)$, which includes all intraparticluar interferences of scattered X-rays, and by the structure factor $S(q)$, which accounts for the interparticular interference. The latter can be neglected in the present case because we are dealing with very dilute solutions of growing ACC particles. The constant K_{SAXS} is the contrast factor in X-ray scattering. Intercepts and initial slopes are established by means of a Guinier fit[15] to the scattering curves in the regime of $0.0011\ nm^{-2} < q^2 < 0.00356\ nm^{-2}$ following

$$\log\left\{\frac{I(q)}{K_{SAXS} c}\right\} \propto \log\{M_w\} - \left[\frac{R_g^2}{3} q^2\right] \tag{3}$$

(11) Faatz, M.; Gröhn, F.; Wegner, G. *Adv. Mater.* **2004**, *16*, 996–1000.
(12) Gebauer, D.; Völkel, A.; Cölfen, H. *Science* **2008**, *322*, 1819–1822.
(13) Porod, G. *Kolloid Z.* **1951**, *124*, 83−114. See also Spalla, O. In *Neutrons, X-rays and Light Scattering Methods Applied to Soft Condensed Matter*; Lindner, P., Zemb, Th., Eds.; Elsevier: Amsterdam, 2002, pp 49−71.
(14) Panine, P.; Finet, S.; Weiss, T. M.; Narayanan, T. *Adv. Colloid Interface Sci.* **2006**, *127*, 9–18.
(15) Guinier, A. ; Fournet, G. *Small-Angle Scattering of X-rays*; Wiley: New York, 1955; Chapter 4.1.1.

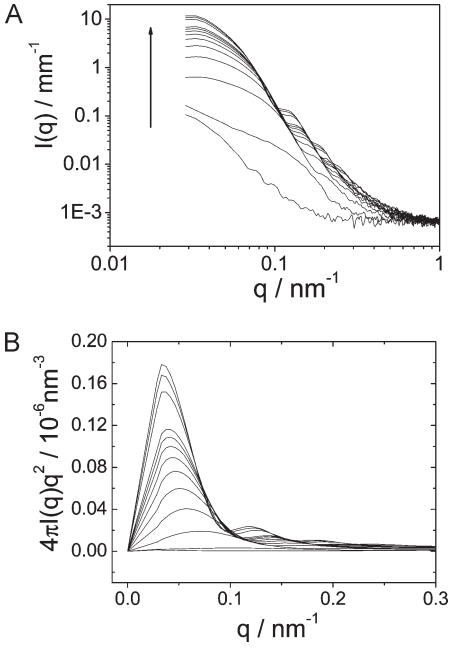


Figure 1. (A) Scattering curves of experiment WOP37 as the differential scattering cross section vs q in the time regime of 0 s $\leq t \leq$ 33 s. The curves from the bottom to the top indicate the increasing age of the particles: 0.04, 0.21, 0.386, 0.567, 0.755, 0.948, 1.148, 1.356, 1.570, 1.792, 4.484, 8.325, and 13.87 s. WOP37 is an example of ACC formation at an initial concentration of 5 mM $CaCO_3$. (B) Scattering curves of experiment WOP37 represented as Kratky plots with $4\pi q^2 I(q)$ vs q for the determination of the invariant Q_p according to eq 12. The curves are based on the same data used in A.

Initial slopes of the respective scattering curves are proportional to the z-average squared radii of gyration, R_g^2.

On the Benefit of the Invariant

Scattering curves of suspensions of homogeneous particles in liquids can be interpreted as the scattering signal of a two-phase system corresponding to a suspension of particles with particle volumes V_p, a volume fraction Φ_p, a density−density correlation function $\gamma_0(r)$, and an average squared excess density of the scattering length $(\Delta\rho)^2$ of the particles with respect to the surrounding medium.

$$I(q) = 4\pi\Phi_p(1-\Phi_p)(\Delta\rho)^2 \int_0^\infty r^2\gamma_0(r)\frac{\sin(qr)}{qr}\,dr \qquad (4)$$

The interpretation of such scattering curves in the limit of dilute solutions is based on a rearranged form of eq 4 with $\Phi_p(1-\Phi_p) \cong \Phi_p$

$$I(q) \cong \Phi_p V_p(\Delta\rho)^2\, P(q) \qquad (5)$$

and with the form factor $P(q)$

$$P(q) = 4\pi \int_0^\infty r^2\gamma_0(r)\,\frac{\sin(qr)}{qr}\,dr \qquad (6)$$

The intercept of the scattering curves is $I(q=0) = K_{SAXS} c M_w$. Expressed by means of eq 5, it reads

$$I(q=0) \cong \Phi_p V_p(\Delta\rho)^2 = \frac{N}{V} V_p^2(\Delta\rho)^2 \qquad (7)$$

with $NV_p/V = \Phi_p$ where N/V is the number density of the scattering particles, each with a volume of V_p.

Equations 4−7 hold only for ensembles of particles with a negligible particle size distribution. This is a reasonable assumption for a monomer addition mechanism with a short nucleation burst compared to the duration of the growth period. In proceeding from equations 4−7, we first consider this possible pathway for particle growth.

A monomer addition mechanism with a short burst of nucleation leads to a binary system with one fraction corresponding to the weakly scattering monomers and the other fraction corresponding to the growing particles. This monomer may correspond to the constituting ions or to a molecular $CaCO_3$ unit or to a cluster of $CaCO_3$ units. However, the exact specification of the monomers is not relevant in the present context as long as this monomer remains a weak scatterer with a negligible intensity contribution. The implications of this feature are outlined in the following paragraph.

If both species (i.e., monomers and particles) have the same scattering length density, then the scattering curve is approximately

$$I(q) \cong (\Phi_m V_m + \Phi_p V_p P(q))(\Delta\rho)^2 \qquad (8)$$

with index m indicating monomers and index p indicating particles. The resulting intercept reads

$$\begin{aligned} I(q=0) &= (\Phi_m V_m + \Phi_p V_p)(\Delta\rho)^2 \\ &= K_{SAXS}(c_m M_m + c_p M_p) = K_{SAXS} c M_w \end{aligned} \qquad (9)$$

In going from eq 2 to eqs −5, we have replaced cM_w by V_pN/V and thus have tacitly neglected the scattering contribution of nonreacted monomers. Such neglect becomes possible for a binary system composed of monomers and particles well separated in size from the monomers. This shall be illustrated by means of an example: An intermediate state along the path to particle formation may include monomers of size $R_m \approx 1$ nm and particles of size $R_p \approx 30$ nm corresponding roughly to a degree of polymerization DP $\approx (R_p/R_m)^3 \approx 30\,000$. If the extent of reaction for this intermediate state is $\Phi_p \approx 0.01$ corresponding to 1% monomer conversion to particles, then 99% of the monomers are left over and the forward scattering is $I(q=0) \approx 0.99 \times 1 + 0.01 \times 30\,000$. Hence, the scattering contribution from the monomers is 0.3% and can be neglected from this state on. In all experiments presented here, a flat scattering of $I \approx 10^{-3}$ mm^{-1} occurred at $q >$ 0.6 nm^{-1}, which is 2 orders of magnitude lower than the forward scattering of even the very first intermediate state analyzed for each kinetic run. Given that this low scattering level stems from monomers, we can safely approximate

$$I(q=0) = K_{SAXS} c M_w \cong K_{SAXS} c_p M_p \qquad (10)$$

because the forward signal $I(q=0)$ of the monomers + particles is approximately equal to the signal stemming from the particles only. If this $I(q=0)$ is divided by the initial concentration c, then the resulting ratio is a weight-average mass M_w, which refers to the total ensemble including strongly scattering growing particles and negligibly scattering monomers. If the same signal is divided by the concentration of the ACC particles c_p (with $c_p = c - c_m$), then the resulting ratio refers to the weight-average mass M_p of the particles only, which is larger than M_w. Although both M values stem from the same signal ($I(q=0)$), they have different physical meanings. In terms of the right-hand side of eqs 7 and 10, an increase in the scattering intensity for a monomer addition mechanism has to be attributed to the growth of the particle volume V_p and, if nucleation is not restricted to a short initial time period, also to an increase in the number of particles N. In the $K_{SAXS}\ cM_w$ term in eq10, the same increase in the scattering

intensity is buried in the conversion of monomers into particles. Once the concentration of particles c_p is known as a value apart from c, there is independent access to the (average) mass M_p of the growing particles in such a case. As will be shown below, this holds significant consequences for the correlation of the size with the mass of the growing particles.

A correlation of the particle size in terms of the square root of the z-average mean square radius of gyration (R_g) with the apparent weight-average molar mass of the particles (M_p) reveals a power law

$$R_g = KM_p^a \tag{11}$$

if these particles have a self-similar structure and do not change the size distribution with increasing average values for M_p and R_g.[16] For densely packed spheres, Gaussian polymer coils, and infinitely thin, rigid rods, the exponent a is $^1/_3$, $^1/_2$, and 1, respectively. However, one has to be careful in using data from light scattering in eq 11 recorded during particle growth by monomer addition.[16] As has been outlined above, such recordings are performed on an ensemble, which includes a decreasing number of monomers. This hardly affects the z-average squared radius of gyration R_g^2, but it has a considerable influence on the weight average of the particle mass M_w accessible via eq 10. It could be demonstrated that the exponents resulting from the growth of self-similar structures following a monomer addition process are just half of the above-mentioned values if M_w instead of M_p is used in eq 11.[16] For spheres, the corresponding exponent is thus expected to be $a = {}^1/_6$. In this sense, SAXS data bear essentially the same uncertainty as the data extracted from TR-SLS.[16] However, resolution becomes possible for the SAXS data for the following reason.

Because of the extended width of the q regime accessible to the SAXS detector, the present TR-SAXS experiments enable an evaluation of the Porod invariant.[13] The Porod invariant corresponds to the total scattering cross section of the sample received by integration over the entire q space according to $Q = \int I(q)4\pi q^2 \, dq$. As can be seen in Figure 1B, the initial part of the invariant was missed for $q < ql$ by the present SAXS setup but it could be easily and accurately estimated by the area of a triangle $\Delta Q_{q<ql} = I(q = ql)4\pi q^2(ql)/2$. Having incorporated this triangle, the invariant Q used in the present work was calculated by integrating curves like those in Figure 1B to the upper limit of $q = 0.3\ nm^{-1}$. Truncation at this limit is justified as follows. A (second) parabolic upturn not shown in Figure 1B becomes noticeable for $q > 0.6\ nm^{-1}$, which can be attributed to background scattering from constituent ions and/or monomers. Thus, the parabolic upturn due to the monomers is noticeable well beyond the q regime where the first Kratky peak attributed to the particles has entirely decayed because of the q^{-4} power law. In fact, the contribution of this peak is already negligible at $q > 0.3\ nm^{-1}$. Hence the Kratky peak from growing particles is separable from an (anticipated) second peak that was attributed to monomers, and the calculated Q_p

$$Q_p = \int_q I(q)4\pi q^2 \, dq = (2\pi)^3 \Phi_p(1-\Phi_p)(\Delta\rho)^2 = (2\pi)^3 \frac{N}{V} V_p(\Delta\rho)^2 \tag{12}$$

is proportional to the actual concentration of scattering particles c_p. As a consequence of the validity of eq 10 (and eq 7), the ratio of the intercept and the invariant makes the particle volume V_p directly accessible via

$$V_p = \frac{(2\pi)^3 I(q=0)}{Q_p} \tag{13}$$

and is proportional to the molar mass of the scattered particles (excluding monomers)

$$M_p \propto \frac{I(q=0)}{Q_p} \tag{14}$$

As already mentioned, the approximation in eqs 10, 13, and 14 is possible as soon as the contribution of the monomers to $I(q = 0)$ becomes negligible. In the present work, this is valid from the very first recording of a scattering curve at 0.04 s.

Along the same line of thought, the number density of the growing particles becomes accessible without any model assumptions of the particle shape. This can be achieved simply by

$$\frac{Q_p}{I(q=0)/Q_p} = \frac{(2\pi)^3 Q_p}{V_p} = (2\pi)^6 \frac{N}{V}(\Delta\rho)^2 \tag{15}$$

leading to a value that differs from the number density of particles only by $(2\pi)^6$ times the average squared excess electron density of ACC.

A monomer addition mechanism does not necessarily lead to particles with a monodisperse size distribution. For systems where the fraction of particles is in itself a polydisperse ensemble, the molar mass of the particles and their particle volume are weight-average values. This makes eqs 8−10 and 13−15 more complicated because the contribution of the particles in eq 8 has to be expressed as a sum according to

$$I(q) \cong (\Phi_m V_m + \sum_i \Phi_{pi} V_{pi} P(q)_i)(\Delta\rho)^2 \tag{16}$$

which includes all size fractions indicated by an index i. This sum can be expressed in terms of averages, which refer to the particles excluding the monomer fraction[13]

$$I(q=0) \cong (\Phi_m V_m + \Phi_p V_{pw} P(q)_{pz})(\Delta\rho)^2 \tag{17}$$

and correspond to the weight average of the particle volume V_{pw} and the z average of the particle form factor $P(q)_{pz}$. Comparison with eq 9 shows that in eq 17 V_{pw} replaces V_P and $P(q)_{pz}$ replaces $P(q)$. Fortunately, averaging does not modify the discussion of exponents in eq 11 for a monomer addition mechanism as long as the polydispersity does not vary with the growth of the particles because the respective average values in eq 11 scale with the same exponent. Thus an exponent of $a/2$ is revealed if R_g is plotted versus M_w derived from eq 9, and an exponent of a is established if M_p from eq 14 has been used instead.

In a particle−particle coalescence mechanism, any solute species, be it a cluster or a particle, may react with any other solute species encountered during Brownian motion. The resulting particle size distribution is broad but monomodal, and the separation of monomers from particles becomes meaningless because the monomers have become part of this monomodal distribution and the distinction between $K_{SAXS}cM_w$ and $K_{SAXS}c_pM_p$ in eq 10 becomes obsolete. Now the intercept of the scattering curves inevitably results in the weight-average particle mass M_w including all species. The invariant of the scattering curves in this case also cannot be subdivided into different contributions and, after all, is no longer needed for the calculation of M_w because c is constant and known beforehand from the sample weight. Along with this, the correlation of the average particle size with the average particle

(16) Liu, J.; Rieger, J.; Huber, K. *Langmuir* **2008**, *24*, 8262–8271.

mass will recover the original exponent, which is $^1/_3$, $^1/_2$, or 1 for a growing sphere, coil, or rod, respectively, if the polydispersity does not change significantly during the coalescence process. This has been demonstrated for spheres growing during a coalescence process according to v. Smoluchowsky.[16,17] The exponent certainly does not change if the intercepts are divided by the (constant) invariant because the meaning of c_p coincides with the constant c in eq 10.

To summarize our discussion, applying eqs 12−15 provides direct access to particle mass (volume) values and particle number densities without any a priori assumptions about the shape of the particles. Above all, the polydispersity in the particle size does not affect the physical significance of eqs 12−15 even if the width of the size distribution varies with time. A monomer addition mechanism with a short initial burst of nucleation can be unambiguously proven by a correlation of the particle mass versus the particle size and by the evolution of the number density of particles with time. If the (average) mass values are extracted from the right-hand side of eq 9, then its correlation to R_g leads to an exponent that is half of the value extracted from the corresponding correlation where the average particle mass refers to particles without monomers.

In similar studies on the formation of gold nanoparticles, the invariant Q_p has also been used. However, it was applied only to estimate the yield of particles.[18] Time-resolved number densities have been estimated independently with model fits[18,19] where the authors had to assume a model for the shape of the particles and a size distribution.

Discussion of Time-Resolved SAXS Experiments

All curves in Figure 1A exhibit undulations that are characteristic of spherically shaped particles with a narrow size distribution. Scattering curves from the other experiments, not shown, exhibit the same features and trends. The results are therefore comparable to the findings published by Bolze et al.[8] who performed similar experiments in a regime of initial $CaCO_3$ concentration of 3.75 mM $\leq c \leq$ 4.5 mM, which is only slightly below the concentration in the present case.

The trends in ACC particle formation for 5 mM $CaCO_3$ shall be discussed by means of the results from four successive experiments. Figure 2 illustrates the evolution of the apparent particle mass expressed as the intercepts of the scattering curves $I(q = 0)$ (upper graph in Figure 2) and of the particle size expressed as the square root of the z-average mean square radius of gyration R_g (lower graph in Figure 2). The process noticeably comes to a standstill after roughly 30 s, when the size and the apparent mass have reached their final values. The final size values fall in the range of 40 nm $< R_g <$ 50 nm. Recordings beyond 120 s (not shown here) usually result in decreasing intercept and size values because of redissolution and/or sedimentation as observed previously.[9]

Figures 3 and 4 present an overview of the time-resolved evolution of the parameters established with eqs 12−15. On the basis of these equations, we may now be able to establish a growth mechanism. First, an exponent of $^1/_6$ for eq 11 is revealed by plotting the radius of gyration of the growing ACC particles versus the bare intercept $I(q = 0) \approx M_w$ corresponding to the weight-average mass value of the ensemble, which includes all solute species. In Figure 5, this correlation is compared to the corresponding correlation of R_g with the particle weight, expressed as

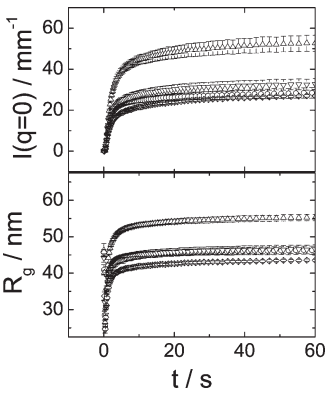


Figure 2. Growth curves for four different formation experiments as the apparent molar mass $I(q = 0)$ (upper plot) and the radius of gyration R_g (lower plot) vs time: WOP34 (○), WOP35 (△), WOP36 (▽), and WOP37 (◇).The initial $CaCO_3$ concentration is 5 mM.

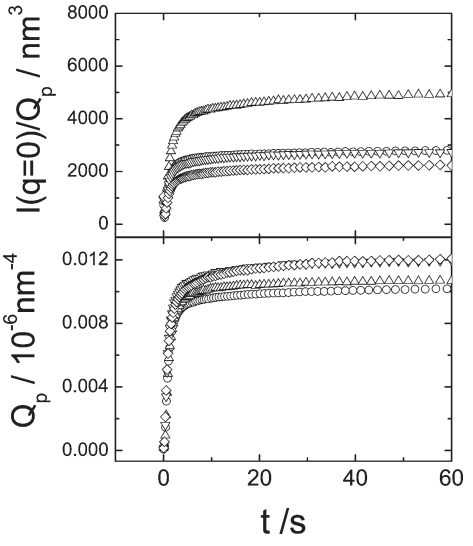


Figure 3. Prorod invariant Q_p (lower plot) and ratio $I(q = 0)/Q_p$ of the intercept and the invariant (upper plot) vs time t for four different formation experiments: WOP34 (○), WOP 35 (△), WOP36 (▽), and WOP37 (◇). The invariant Q_p is proportional to the amount of $CaCO_3$, converted into ACC partcles. The ratio $I(q = 0)/Q_p$ is proportional to the weight-average particle volume V_p. The symbols refer to the same experiments as in Figure 2.

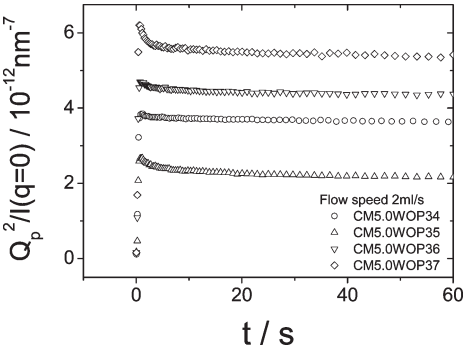


Figure 4. Ratio $Q_p^2/I(q = 0)$ as a parameter representing the number densities of particles vs time t for four different formation experiments: WOP34 (○), WOP 35 (△), WOP36 (▽), and WOP37 (◇). The symbols refer to the same experiments as in Figure 2.

$M_p \approx I(q = 0)/Q_p$, which is in line with eqs 13 and 14. In the latter case, an exponent of $^1/_3$ can be recovered from the correlation,

(17) v. Smoluchowski, M. *Phys. Z.* **1916**, *17*, 585–599.
(18) Abécassis, B.; Testard, F.; Spalla, O.; Barboux, P. *Nano Lett.* **2007**, *7*, 1723–1727.
(19) Polte, J.; Ahner, T. T.; Delissen, F.; Sokolov, S.; Emmerling, F.; Thünemann, A. F.; Kraehnert, R. *J. Am. Chem. Soc.* **2010**, *132*, 1296–1301.

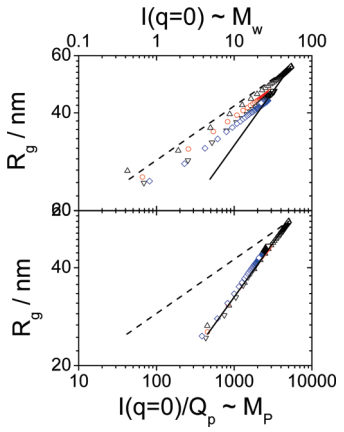


Figure 5. Correlation of the particle size expressed as the radius of gyration with particle mass data. Mass data are represented as the intercepts $I(q = 0)$ of the scattering curves (upper plot) and as the ratio $I(q = 0)/Q_p$ (lower plot). The straight lines indicate exponents of $a = {}^1/_6$ (---) and ${}^1/_3$ (−). The symbols denote the experiment: WOP34 (red ○), WOP 35 (△), WOP36 (▽), and WOP37 (blue ◊). Intercepts are considered to be proportional to M_w, and the ratio $I(q = 0)/Q_p$ is proportional to the weight-average particle mass M_p.

confirming the spherelike shape of the growing ACC particles. It is worth mentioning here that the data do overlay much better once R_g is plotted versus $I(q = 0)/Q_p \approx M_p$ instead of $I(q = 0) \approx M_w$. This not only indicates the consistency of the procedure suggested by eqs 11−15 but also proves that the particles in all four experiments have the same density and polydispersity. More importantly, the fact that the two exponents differ by a factor of 2 clearly proves the growth of a monodisperse fraction of ACC particles according to a monomer addition mechanism.

From an alternative perspective, the appearance of two exponents a and ${}^a/_2$ in eq11 is an inevitable consequence of a nucleation burst, which generates a constant number density N/V of growing particles. In such a case, we have $I(q = 0) \approx (N/V)R_g^6$ and $Q_p \approx (N/V)R_g^3$ and thus $I(q = 0)/Q_p$ scales with R_g^3. Given that the contribution from the particles and the monomeric units to Q_p are well separable and we succeeded in capturing exclusively Q_p of the particle fraction, the time-resolved evolution of $Q_p/[I(q = 0)/Q_p]$ should recover a constant number density of particles if our data evaluation is consistent. As demonstrated in Figure 4, this evolution indicates a sharp increase in the particle number density, which is accomplished after 1 s and corresponds to the nucleation step. The successive plateau value seems to follow a small maximum, which at present cannot be safely considered to be significant. The growth period of roughly 20−30 s extends over a period where $Q_p/[I(q = 0)/Q_p]$ is already constant. This growth period is considerably longer than the nucleation period. As expected, it is this sequence of steps and its relative durations that are responsible for the formation of monodisperse particles. The duration of both steps is expected to depend on the ACC concentration and on the mixing technique. These results nicely reveal a particle formation mechanism where particles grow as a uniform sample immediately after a short burst of particle nucleation.

At this point, a comment on the spread of the absolute values of the number density is of interest. Despite this spread, the extents of conversion, reflected in the trends in Q_p, are surprisingly close to each other. Apparently, it is the rate of nucleation that responds the most sensitively to any changes in environmental conditions. A variation in the number of nuclei is compensated for by corresponding changes in the final particle size values.

It has to be emphasized that our results help to reconsider the growth mechanism of the ACC particles, which had to remain open in a paper published recently by Cölfen et al.[12] Cölfen et al.[12] found clear evidence for an equilibrium between neutral $CaCO_3$ clusters and constituent ions during the prenucleation stage. These clusters turn into nuclei, once they achieve a critical size. With our results regarding the exponents in eq 11 and the number of growing particles, a particle−particle coalescence pathway can be excluded as the particle growth mechanism. Instead, a short burst of nucleation generates a number of nuclei and hence a number of growing particles. The particles grow at the expense of monomers whereby the number of particles remains constant. The mechanism is denoted a monomer-addition process and closely resembles the model suggested by LaMer and Dinegar for the formation of monodisperse hydrosols.[20] Like Cölfen et al.,[12] we cannot further specify the type of monomer (i.e., whether it corresponds to clusters or constituting ions), but in light of the results from Cölfen et al.[12] and from Pouget et al.,[21] we are inclined to attribute the (prenucleation) clusters identified by those authors[12,21] to the species that acts as the monomer in the monomer-addition mechanism established presently for particle growth. However, a comparison of the present results with the work by Cölfen et al.[12] and Pouget et al.[21] has to be considered with great caution because the way that they induced cluster formation differs significantly from the rapid mixing with a stopped-flow device. In the study by Cölfen et al.,[12] supersaturation was approached stepwise. Each step addressed a distinct and equilibrated prenucleation state, and in these steps, evidence is provided for (prenucleation) clusters. In contrast, our stopped-flow experimentation induced nucleation almost instantly. The rapid mixing generates turbulences and, along with it, local inhomogeneities in the degree of supersaturation, all of which may influence nucleation,[7] and we simply do not know whether those prenucleation clusters also occur under conditions induced with the stopped-flow device.

Similar conclusions could have been inferred from the data published by Bolze et al.[8] They established particle volumes by fitting the polydisperse sphere model to the scattering curves and revealed very low degrees of polydispersity and growing average particle volumes, which approach a plateau value within a few tens of seconds. However, the present analysis along the lines of eqs 11−15 achieves a clear picture of the particle formation mechanism without any model assumptions made about the particle shape and thus underlines the power of time-resolved SAXS analysis.

A more quantitative interpretation of the present data demonstrates the consistency of the analysis suggested by eqs 11−15. To this end, we infer an excess electron density of the ACC particles of 184 electrons per nm^3 from Bolze et al.[8] and apply the electron differential scattering cross section of $\sigma_0 = 7.94 \times 10^{-24}$ mm^2 in order to reach the following average square excess density of the scattering length for the ACC particles

$$(\Delta\rho)^2 = 0.26910^{-6}(nm^{-4}) \quad (18)$$

The use of this value in eqs 12 and 15 enables the calculation of particle volume fractions and the number densities of particles in mm^{-3}. In addition, absolute values for the weight-average particle volumes become accessible via eq 13. The results are summarized in Table 1. A comparison of the data of all four runs shows

(20) LaMer, V. K.; Dinegar, R. H. *J. Am. Chem. Soc.* **1950**, *72*, 4847–4854.
(21) Pouget, E. M.; Bomans, P. H. H.; Goos, J. A. C. M.; Frederik, P. M.; de With, G.; Sommerdijk, N. A. J. M. *Science* **2009**, *323*, 1455–1458.

Table 1. Limiting Size Parameters and Number Densities of the ACC Particles from Formation Experiments with an initial concentration of 5 mM $CaCO_3$ and with a Flow Rate of 2 mL/s[a]

sample	a	N/V (10^8 mm^{-3})[b]	R_g (nm)	V_p (10^6 nm^3)[c]	$V(R_g)$ (nm^3)[d]	Φ_p (10^{-4})[e]
5.0WOP34	0.328	2.19	46.55	0.706	0.909	1.55
5.0WOP35	0.334	1.30	55.48	1.238	1.539	1.61
5.0WOP36	0.374	2.54	46.73	0.684	0.920	1.74
5.0WOP37	0.330	3.26	43.57	0.556	0.745	1.81

[a] The volume fractions of ACC particles in the last column have to be compared with a volume fraction of $\Phi_p = 3.58 \times 10^{-4}$ corresponding to a 100% conversion. (See the text for further details.). [b] Calculated with eqs 15 and 18. [c] Calculated with eq 13. [d] Calculated with eq 19. [e] Calculated with eqs 12 and 18.

that the plateau values reached for the particle volumes and the number densities only qualitatively agree. It turns out to be impossible to reproduce these numbers quantitatively in going from one experiment to the next. One reason for this fluctuation may be the extreme sensitivity of the nucleation to pH values in particular. However, this does not affect the essence of our quantitative results, which have the following features. (i) The particle number densities approach a few 10^8 per mm^3. This is significantly larger than the number established by Bolze et al.[8] (1.75×10^7 mm^{-3}) at an initial $CaCO_3$ concentration of 4.5 mM. The discrepancy may easily be attributed to the fact that the present values are determined at a slightly higher initial calcium carbonate concentration of 5 mM compared to 4.5 mM investigated by Bolze et al.[8] Although this is only an increase in the concentration by 10%, the degree of supersaturation of ACC is roughly increased by a factor of 1.2 when the solubility of ACC is 1.7 mM.[4] An even more drastic increase in the number density by a factor of 1000 is also reported[10] once the concentration is increased by a factor of 2.2 from 4.5 to 10 mM, which underlines the sensitivity of nucleation toward the initial $CaCO_3$ concentration. (ii) Despite the large difference in the respective number densities reported in the present work and in ref 8, it seems that the amount of ACC being formed is again close to that in both cases. This becomes transparent by looking at the final particle size values. The particles observed in ref 8 approach a final radius of 130 nm at 4.5 mM and are larger by a factor of 2 to 3 compared to our final values. An increase in the radius by a factor of 2 increases the mass by a factor of 8, which is nicely compensated for by the number of particles decreasing by at least $^1/_6$. (iii) The weight-average volumes V_p derived from the intercept according to eq 13 agree well with the corresponding value calculated with the z-average radius of gyration according to

$$V(R_g) = {R_g}^3 \left(\frac{5}{3}\right)^{1.5} \frac{4\pi}{3} \tag{19}$$

The values of $V(R_g)$ are persistently larger than V_p, in line with the fact that z averages of ${R_g}^2$ have a higher weighting on the large particles than does the weight average of V_p with respect to the latter, corresponding to a lower moment than the z averages. (iv) Table 1 includes volume fractions of ACC particles after 60 s for all four runs, calculated according to eq 12 and 18. The volume fraction corresponding to $Q_p = 0.011 \times 10^{-6}$ nm^{-4} averaged at 60 s is roughly 1.65×10^{-4}. This value is in very good agreement with the value found by Bolze et al. at 4.5 mM $CaCO_3$.[8] To relate this value adequately to a 100% conversion of 5 mM $CaCO_3$ to ACC particles, we first need the mass density compatible with the respective electronic density (in eq 18), inferred from Bolze et al.[8] They estimated a 1.62 g/cm^3 value for the mass density of ACC. This density has to be multiplied by the mass of 5 mM $CaCO_3$, including the mass of the amount of water being incorporated into ACC by this 5 mM $CaCO_3$. If we follow the findings by Brecevic and Nielsen[4] and assume that 1 water molecule per $CaCO_3$ unit is incorporated into ACC, then a volume fraction of 3.6×10^{-4} is recovered after the complete conversion of $CaCO_3$ units to ACC particles. The latter number corresponds to an ACC solubility of 2.7 mM, which is surprisingly close to the level of 1.7 mM estimated by Brecevic and Nielsen.[4]

The knowledge established for the shape of the growing ACC particles suggests a direct interpretation of the data with model curves for polydisperse spheres. To this end, a fit has been used first according to eq 16 by neglecting the contribution $\Phi_m V_m$ of the remaining monomeric units. The Rayleigh formula[22] for homogeneous spherical particles was used for the particle form factor $P(q)$. The distribution of the particle size, expressed by $\Phi(V)$ is assumed to follow a Schulz−Zimm distribution.[23] The width of the distribution is defined by the ratio of the weight-average volume divided by the number-average volume of the particles V_{pw}/V_{pn}, which can be translated into the Schulz−Zimm polydispersity parameter z according to

$$z = \frac{1}{V_{pw}/(V_{nw} - 1)} \tag{20}$$

Fits were performed to the data from experiment WOP37. The q regime selected covered the range of $0.0685\ \text{nm}^{-1} < q < 0.1347\ \text{nm}^{-1}$, which includes the first undulation of the experimental scattering curve. The fits revealed the same size versus mass correlation as that extracted from the Guinier analysis used in Figure 4. However, a closer look at these fits indicated that the model fits exhibited a slightly steeper overall decay than did the experimental curves, predominantly at a low extent of reaction. This may be due to unreacted constituent clusters or building units, which contribute as a second component with additional and almost flat scattering. To improve the model fits, a second contribution has been incorporated into the sphere system by means of an additive Lorentzian scattering function[24] for random fluctuations. The fluctuations have a correlation length ξ that is significantly smaller than the size of the spherical particles themselves.

$$S(q) = \frac{S(0)}{[1 + (q\xi)^2]} \tag{21}$$

As is demonstrated in Figure 6, a satisfactory fit can indeed be achieved now for the complete scattering curve covering the regime of $0.029\ \text{nm}^{-1} < q < 1.592\ \text{nm}^{-1}$ with such a bimodal system. We again consider this to be a hint for the scattering contribution of the remaining $CaCO_3$ building units not yet incorporated into the growing ACC particles.

Summary

The formation of ACC particles was induced at an initial concentration of 5 mM $CaCO_3$ with a stopped-flow device by mixing the respective component solutions. The process came to a standstill after 30 s, where metastable spherical particles with a size of 40−50 nm were formed. SAXS curves gave access to the Porod invariant Q_p, which could consistently be attributed to the amount of $CaCO_3$ incorporated into particles. It is this feature of the invariant that enabled the calculation of the mass values M_p and the number densities N/V of the growing particles without any assumption of a model for the particle shape. The number

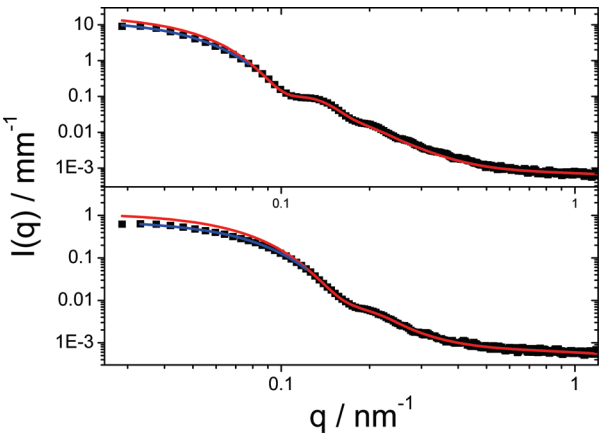


Figure 6. Model fit with form factors for polydisperse spheres[22,23] including an additional decay from a Lorentzian term.[24] The two scattering curves belong to experiment WOP37 and refer to t = 0.386 s (bottom) and t = 3.868 s (top). The curves indicate the following fits: a Guinier fit to the low-q regime (blue −) and a model fit in the regime of momentum transfer of 0.029 nm^{-1} < q < 1.592 nm^{-1} (red −). The fit parameters are ξ = 0.45 nm, R_g = 19.2 nm, and V_{pw}/V_{pn} = 1.21 (0.386 s) and ξ = 0.35 nm, R_g = 31.9 nm, and V_{pw}/V_{pn} = 1.17 (3.868 s).

density of particles approached their final values within 1 s. Successive growth proceeded by an increase in particle size and an increase in the particle weight fraction while the number of particles remained constant and the size distribution of the particles remained narrow. This is compatible with a monomer addition mechanism being composed of two steps (i.e., a short nucleation step followed by a long growth period). Nothing can be said about the nature of the monomers, which may correspond to constituting ions or just as well to clusters. However, if the clusters are the monomers, then the system has bifurcated by the time the process reaches the turn from the first to the second step: clusters either became nuclei and acted like growing particles or remained as such and acted like monomers. Confirmation of this monomer addition mechanism as well as the indication of the spherical shape of the growing particles was provided by establishing a correlation between the particle size R_g and the particle mass. As expected for a monomer addition mechanism of spheres,[16] this correlation leads to an exponent of $^1/_6$ if the weight-average solute mass M_w is used and an exponent of $^1/_3$ if the weight-average particle mass M_p is used instead. A particle−particle coalescence mechanism can be ruled out by our results because such a mechanism would induce a decreasing number of particles and does not allow for the discrimination of M_p and M_w because the particle size distribution is not bimodal with two rather monodisperse fractions but rather is broadly monomodal.

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(22) Rayleigh, L. *Proc. R. Soc.* **1914**, *A90*, 219–225.
(23) Zimm, B. *J. Chem. Phys.* **1948**, *16*, 1093−1099. Zimm, B. *J. Chem. Phys.* **1948**, *16*, 1099−1116.
(24) Ornstein, L. S.; Zernike, F. *Proc. Sect. Sci. K. Med. Akad. Wet.* **1914**, *17*, 793.